PYTHON ACTIVITY-BASED LEARNING

README • Project Documentation • Academic Year 2026–2027

Project Overview

This repository contains the Python Programming Activity-Based Learning (ABL) assignment for the academic year **2026–2027**. The consolidated report presents 10 selected case studies. Each case study includes a problem statement, objectives, assumptions, algorithm, flowchart, pseudocode, Python source code, test cases, sample-output screenshot, and conclusion, following the assignment circular.

Project Objectives

- Understand practical problems using Python.
- Develop logical and computational thinking.
- Represent solutions using algorithms and flowcharts.
- Implement and test Python programs.
- Analyse program outputs and draw conclusions.

Selected Case Studies

The consolidated report contains the following selected case studies and titles:

1.Case Study 12 — Attendance Recovery Computation
Calculates current attendance and the minimum number of additional fully attended classes needed to reach the 75% eligibility threshold.

2.case study ATM Transaction Sequence Simulation
Simulate a sequence of deposits and withdrawals, reject invalid transactions, and report the final balance and transaction counts.

3. Multi-Product Shopping Budget Constraint
Calculate the total cost of a basket of products, apply a threshold discount, and identify whether the basket fits the budget.

4. Repeated Transaction Detection
Inspect transaction IDs and amounts, identify repeated IDs, and flag transactions whose amount exceeds a supplied limit.

5. 5. Password Policy Validation Engine
Check a sample password against a supplied minimum length and character-category rules, and explain which rules fail.

6. Student Feedback Sentiment Scoring
Score multiple feedback comments using a supplied positive and negative keyword list and classify each comment.

7. Text Compression Using Character Runs
Encode consecutive repeated characters as character-plus-count groups and use the encoded version only when it is shorter

8. Student Marks Matrix Analysis
Analyse marks for several students across multiple subjects, calculate each student average and each subject average, and identify the highest student average.

9. Weekly Electricity Consumption Matrix
Analyse appliance-wise electricity consumption over seven days, calculate daily and appliance totals, and flag daily usage over a supplied limit.

10. Food Waste Monitoring System
Compare meals prepared, served, and left over across multiple days, calculate wastage percentage, and flag days exceeding a supplied threshold.

Python-Activity-Based-Learning-Decorated consolidated ABL report containing the 10 selected case studies, Python

programs, test cases, sample-output screenshots, and flowcharts.

If the PDF is renamed in the repository, update the filename above so that it exactly matches the uploaded file.

Tools Used

- **Python** — program implementation and testing
- **GitHub** — repository hosting and submission
- **Markdown** — project documentation

Conclusion

This ABL project supports the development of Python programming skills, logical problem-solving, algorithm design, flowchart interpretation, and program testing through practical case studies.

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